

Ayush Urs

858-209-0522 | ayushurs11@gmail.com | linkedin.com/in/ayush-urs-515a42205 | github.com/ayush11z | US Citizen

EDUCATION

University of California, San Diego

M.S. Computer Science

San Diego, CA

September 2026 – Present

University of California, San Diego

B.S. Mathematics – Computer Science

San Diego, CA

GPA: 3.89 September 2023 – March 2026

Relevant Coursework: Machine Learning: Algorithms, Data Structures, Stochastic Processes, Recommender Systems

TECHNICAL SKILLS

Languages: Java, Python, C/C++, SQL, JavaScript, Dart, MATLAB, Stata

Frameworks: Apache Spark, Databricks, Spark SQL, TensorFlow, PyTorch, Keras, LangChain, FastAPI, Flutter

Tools: PostgreSQL, MongoDB, SQLite, Docker, Git, Google Cloud Platform, Vertex AI, Nautilus, VizTracer, HPC, TPU/V100 GPU, Jupyter, VS Code, Cursor, LoRA, MCP(Model Context Protocol)

EXPERIENCE

Student Researcher

Remote

University of Wisconsin–Madison

June 2026 – Present

- Designing multi-agent **LangChain** pipeline routing agents for literature parsing, code generation, and debugging to translate papers into parameterized **PhysiCell/Morpheus** simulations, grounding sweeps in **NCBI** literature.
- Extending a **RAG** retrieval layer over simulation-library Java API docs and reference manuals to improve code-generation accuracy, and scoping an **MCP server** to standardize tool access across the agent pipeline.

ML Research Assistant

San Diego, CA

Halcioğlu Data Science Institute

March 2026 – June 2026

- Extended **SkyDiscover** (EvoX co-evolution framework) implementing **EoH** and **FunSearch**-style search strategies with **bandit-based selection** and **divide-and-conquer decomposition**, evolving solution programs and search heuristics via fitness-scored selection over a **SQLite** strategy database.
- Benchmarked **LLM-driven algorithm design** pipelines, developing **unit tests** and using **VizTracer** for real-time execution profiling across optimization runs.

Machine Learning Researcher

San Diego, CA

Human-centered eXtended Intelligence (UCSD Lab)

September 2025 – Present

- Built scalable data pipelines for LLM alignment using Reddit APIs and cloud infrastructure (**Vertex AI, GCP**), collecting and preprocessing **100K+ emotion-regulation samples** structured across **CERQ dimensions**.
- Fine-tuned **transformer models** using **Direct Preference Optimization (DPO)** and **LoRA** on multi-GPU Nautilus clusters, improving preference alignment and sensitivity to human emotional cues.

Research Assistant — Parkinson's Voice Detection

New York, NY (Hybrid)

Myna Voice Labs (Research Collaboration with Princeton PhD Scholars)

August 2025 – April 2026

- Developing AI models for Parkinson's disease detection from speech using **SVMs, CNNs, and Audio Spectrogram Transformers (AST)**, evaluating model robustness under audio perturbations and noise, achieving **89% accuracy** with strong cross-linguistic generalization.
- Optimizing audio feature extraction (**MFCC, OpenL3, Whisper**) and exploring **LoRA-based continual learning** to mitigate catastrophic forgetting in large transformer models (**~900M parameters**).

PROJECTS

CNN Skin Lesion Detection System | *TensorFlow, Flutter, Python, HAM10000*

- Built a **CNN-based** skin lesion classifier using **TensorFlow** and the **HAM10000** dataset, achieving **82%** classification accuracy for early melanoma detection through advanced **data augmentation** techniques.
- Deployed a **Flutter** mobile application enabling **real-time dermoscopic image analysis** for beta users, integrating the trained model as a live inference backend.

RAG Pipeline for Internal Query Answering | *LangChain, sentence-transformers, MongoDB, FastAPI*

- Built **RAG** pipeline with **LangChain, sentence-transformers**, achieving **75%** relevance across **1K+** queries.
- Designed hybrid data layer with **MongoDB Vector Search** and **PostgreSQL**, orchestrated via **FastAPI** and deployed with **Docker**, sustaining **100+** concurrent requests at **p95 latency of 1–2 seconds**.

AI-Integrated Island for Eutrophication Control | *Reinforcement Learning, Python*

- Developed AI-driven organic island research prototype using Bellman equation to autonomously target high-phosphate regions, achieving **75% phosphate reduction** in lab trials.
- Implemented value-state matrix for intelligent navigation across grid-based lake environments.